1️⃣ Equivalence Partitioning (EP)

Divide possible inputs into **groups/classes** where you expect the system to behave similarly.

Then test **one representative value from each group**.

🔥 **Example: Age**

Requirement:

> User must be between **18 and 60**.

Instead of testing every number, divide the input into partitions:

```
INVALID        VALID          INVALID
─────────┬──────────────┬──────────────
  < 18   |    18–60    |    > 60
─────────┴──────────────┴──────────────
```

So you could select:

Partition	Example	Expected
Invalid	17	Reject
Valid	30	Accept
Invalid	61	Reject

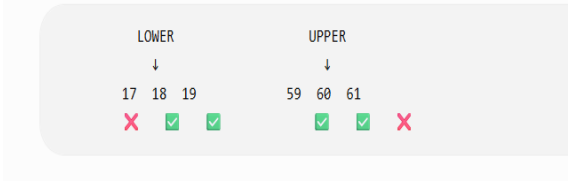
That's **Equivalence Partitioning**.

2️⃣ Boundary Value Analysis (BVA)

BVA says:

Bugs love boundaries.

Visual:



"What happens at the edges of those groups?

3️⃣ Decision Table Testing :

Now things get interesting because different combinations produce different outcomes.

A decision table helps organize this.

Conditions:

Username correct

Password correct

Account active

The output depends on multiple conditions/combinations.

Logged in	Permission	Document exists	Archived	Result
Yes	Yes	Yes	No	Download
Yes	No	Yes	No	Deny
No	Yes	Yes	No	Login required
Yes	Yes	No	No	Not found
Yes	Yes	Yes	Yes	Deny

State Transition Testing

Now we're dealing with states.

The system behaves differently **depending on what happened previously.**

Imagine a bank card.

Initial state:

```
ACTIVE
```

Enter wrong PIN once:

```
ACTIVE
   ↓
1 wrong attempt
   ↓
ACTIVE
```

Second wrong attempt:

```
ACTIVE
   ↓
2 wrong attempts
   ↓
ACTIVE
```

Third wrong attempt:

```
ACTIVE
   ↓
3 wrong attempts
   ↓
BLOCKED
```

Now the system is in:

```
BLOCKED
```

Even if you enter the correct PIN:

```
Correct PIN
      ↓
BLOCKED
      ↓
Transaction denied
```

That's **State Transition Testing.**

When you get a requirement in an interview, mentally ask these four questions:

1. Are there ranges/groups?

👉 **Equivalence Partitioning**

```
GROUPS
```

2. Are there limits?

👉 **Boundary Value Analysis**

```
EDGES
```

3. Are there multiple conditions?

👉 **Decision Table**

```
COMBINATIONS
```

4. Does previous behavior affect current behavior?

👉 **State Transition**

```
HISTORY / STATES
```